



In addition to part I (the General Handout for all courses, appended to the Timetable), this section provides further specific details regarding the Course.

COURSE NO.	:	BITS F314
COURSE TITLE	:	GAME THEORY & ITS APPLICATIONS
INSTRUCTOR-IN-CHARGE	:	PROF. BALAKRUSHNA PADHI
INSTRUCTOR	:	PROF. RAHUL ARORA

1. COURSE DESCRIPTION

This course introduces the fundamental concepts and analytical tools of game theory for studying strategic interactions among rational decision-makers. It covers the construction and classification of games with complete and incomplete Information, strategic behavior, and solution concepts, including dominant strategy equilibrium, iterated elimination of strictly dominated strategies (IESDS), Nash equilibrium, mixed strategy equilibrium, and static and dynamic games with and without complete Information. It also includes extensive-form games, finite- and infinitely-repeated games, Bayesian games, and the integration of linear programming with game-theoretic analysis. Emphasis is placed on applications to economics and public policy, including social choice and voting theory, trade, environmental economics, industrial organization, auctions, negotiations, and other strategic decision-making problems.

2. SCOPE AND OBJECTIVE OF THE COURSE

Game Theory is a tool and is a beautiful mix of mathematics, economics, and psychology. Game theory is a technique for analyzing strategic problems across diverse settings. The foundation of Game Theory was laid by John von Neumann, who proved the fundamental minimax theorem in 1928 and published a book (co-authored with Oskar Morgenstern), *Theory of Games and Economic Behavior*, in 1944. Game Theory became a buzzword and attained particular glamour, with the Nobel Prize going to John Nash for his contributions to the Field, which has found applications in almost every branch of knowledge. Broadly, Game Theory can be divided into cooperative and non-cooperative game theory. This course deals only with non-cooperative games. The main objective is to introduce students to Game Theory & Its applications to the real world as a tool for logical thinking in a strategic interaction setting.

3. COURSE LEARNING OUTCOMES

The course would aim to illustrate economic & other real-world concepts and to make students adept at approaching financial problems from a game-theoretic perspective.

- CLO1)** Understand the Fundamentals of Game Theory, including game construction, analysis, and solutions.
- CLO2)** Able to apply game theory tools to real-world contexts such as economic, political & environmental settings, trade negotiations, environmental agreements, cartels, oligopolistic competition, and in social choice contexts, among others.
- CLO3)** Understand the collective decision-making mechanisms using game-theoretic tools.
- CLO4)** Gain exposure to games with imperfect Information, including understanding the basic concepts of Bayesian games (types, beliefs, and strategies).
- CLO5)** Develop problem-solving and strategic thinking skills.

4. TEXTBOOKS AND REFERENCE BOOKS

4.1 Textbooks

1. Osborne, Martin J. (2004): *An Introduction to Game Theory*, Oxford University Press, India.



4.2 Reference Books

1. Webster, Thomas J. (2009): *Introduction to Game Theory in Business and Economics*, Segment Books.
2. Gibbons, Robert (2019). *Primer in Game Theory*. Pearson.
3. Dutta, P. K. (1999). *Strategies and games: theory and practice*. MIT Press.
4. Fudenberg, Drew & Tirole, Jean (2005): *Game Theory*, MIT Press.
5. Osborne, Martin J. & Rubinstein, Ariel (1994): *A Course in Game Theory*, The MIT Press Cambridge.
6. Dixit, A. K., & Skeath, S. (2015). *Games of strategy*: Fourth international student edition. WW Norton & Company.
7. Dixit, A. K., & Nalebuff, B. (2008). *The art of strategy: a game theorist's guide to success in business & life*. WW Norton & Company.

5. Course Format & Weekly Workflow

The course will be conducted using a blended, student-centered, and outcome-based approach, incorporating regular lectures (three per week to study core concepts and applications through the usage of Blackboard & multimedia tools such as PowerPoint slides and video clippings) and interactive discussions during lectures (to encourage discussion on real-life cases and to promote critical thinking). The students' progress throughout the course will be evaluated through a continuous assessment approach in accordance with the institute's academic regulations. The grading will be relative and in accordance with the BITS academic regulations. The course will use NALANDA, a learning management system (LMS), to display course information, including lecture slides/notes, practice sheets, solution materials, and other related materials. All notices will also be published on NALANDA.

5. COURSE PLAN

Week No.	Module No. [Description]	Description Lecture Session	Reference Material	Learning Outcome
1	1 [Introduction to Game Theory and Basic Concepts]	L1: Basic Introduction to Game Theory; How to form a game; Types of Games & their applications; L2: Prisoner's Dilemma; Examples of Strategic Games; Strictly Dominant Strategy Equilibrium; Defining the concept of Nash Equilibrium (PSNE)	Ch 1 (TB1 and RB1) & Class Notes	Understanding of the history of game theory, theory of rational choice, and Interactive decision making
2-3	2 [Games with complete Information: Solution Concepts for Simultaneously Played finite games with Discrete Strategies, Non-cooperative, One-Time, Static Games]	L3: Strictly & Weakly dominant strategies, Nash equilibrium, Dominant strategy equilibrium (IESDS); L4: Best Response Method, & Arbitrary decision rules (Maximin only), three-player game; L5: Minimax decision rule, Focal point Equilibria, Saddle point calculation for zero-sum games, oddment method for mixing strategies; L6: Minimax theorem, mixing pure strategies through an optimal mixing rule; Dominance with mixed strategies: Strict & Weak (MSNE); L7: Infinitely many equilibria, Odd rule, Free Money Game, Conversion of	Ch 2 & 4 (TB1), Ch 2, 3, 6 (RB1) & Class Notes	Students would be able to model various situations from their day-to-day lives using game-theoretic methods, including many economic applications, and begin reasoning about strategic interaction. They will also be prepared to adopt a mixed strategy to avoid predictability.



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		a game into a linear programming problem, Travelers' Dilemma Game		
4	3 [Infinitely/ Finitely Repeated, Static Games with Complete Information]	L8: Introduction to Infinitely & Finitely Repeated Static Games; L9: Evaluating Payoffs in Infinitely Repeated Games, Trigger Strategies; Cheating Rule for Infinitely Repeated Games, Making Threats Credible L10: Tit-for-Tat strategy, Finitely Repeated Games with a Certain End, Cheating Rule for Finitely Repeated Games	Ch 15 (TB1), Ch 4, 5 (RB1), & Class Notes	Strategic Interaction and Decision-Making in a Dynamic Game
5	4 [Dynamic/Extensive Games with Perfect Information]	L11: Understanding Extensive Games with Perfect Information; Formation of Game Trees and Practical Analysis; L12: Subgame Perfect Equilibrium SPNE; Backward Induction; First Mover Advantage; L13: Conversion of Extensive Form into Normal Form of Game. Multiple SPNEs and Stage Game	Ch 5 (TB1), Ch 11 (RB1), & Class Notes	Strategic Interaction and Decision-Making in a Sequential Game
6-7	5 [Applications of Nash Equilibrium to Economic Markets with Complete Information]	L14: Basics of continuous strategies game, its solution through calculus; Applications to the markets in game theory: Cournot model, theoretical understanding L15: Practical application of the Cournot model through game theory L16: The Stackelberg model of game theory L17: Bertrand model theory and practice in game theory	Ch 3 (TB1), Ch 8, 9 (RB1), & Class Notes	Strategic Interaction and Decision Making among players in different markets
8	6 [Games with complete Information: Solution Concepts for Simultaneously Played finite games with Continuous Strategies]	L18: Applications to Environment: Tragedy of the Commons, The Coase Theorem, Shifting Reaction Functions L19-20: Applications to Trade Policy & Practice Session	Ch 7 & 10 (RB1) & Class Notes	
L21 - Quiz 1 before Mid-Sem – For Schedule, see evaluation scheme				
9	7 [Applications of Game Theory in Social Choice & Others with Complete Information]	L22: Decision making in Electoral Competition; Voting Principles and Voting Paradox L23: Analyzing Median Voter in-game theory L24: Analyzing the War of Attrition in a game-theoretic framework	Ch 3 (TB1), & Class Notes	Strategic interaction and Decision-making among social actors
10-11	8 [Applications of Static & Dynamic Games with	L25: Analyzing Ultimatum & Bargaining Games L26: Ultimatum and bargaining games and their practical applications	Ch 6 (TB1), Ch 12, 15 (RB1),	Analyze the Bargaining Situations with and without Impatience



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	Complete Information]	L27: Introduction to Auction Taxonomy L28: Kind of Auction: understanding Closed Bid & Open Bid auctions L29: Analyzing auctions in static games L30: Analyzing auctions in dynamic games	& Class Notes	
12-14	9 [Static and Dynamic Games with Imperfect Information]	L31: Introduction to game with incomplete Information; Types, beliefs, and common prior assumptions. L32-33: Introduction to Bayesian Nash Equilibrium (BNE): concept and computation; Common Applications of Static BNE; L34-35: Dynamic Games with Incomplete Information; Perfect Bayesian Equilibrium (PBE): concept and components L36-L37: Signaling Games; Basics and Equilibria; Pooling and Separating Strategies L38-L39: Auctions with Independent Private Values; Principal-Agent Problem	Ch 9 (TB1), Ch 16 (RB1), & Class Notes	Introduce the concept of information sets and the use of Bayesian Updating to find a perfect Bayesian Equilibrium.
	10 Master Class	L40-41: <i>It is proposed to conduct one Master Class on Applications of Game Theory by an External Expert post mid-semester examination, and the concepts taught will be a part of the syllabus</i>	Class Notes	Apply game-theoretic concepts to analyze complex strategic interactions and gain insights into their real-world applications through expert-led case discussions.
	L42 – Quiz 2 before the Comprehensive post-Mid-Sem– For Schedule, see evaluation scheme			
	L43 – Quiz 3 before the Comprehensive post-Mid-Sem– For Schedule, see evaluation scheme			

6. EVALUATION SCHEME

Component	Max Duration	Weightage	Max. Marks	Tentative Dates*	Nature of Component
Quiz – 1	45 minutes	15%	30	23 Sept 2026	Closed Book
Mid-Semester Examination	90 minutes	30%	60	06 Oct 2026 (AN1)	Open Book
Quiz – 2	45 minutes	15%	30	06 Nov 2026	Closed Book
Quiz – 3[#]	45 minutes	15%	30	23 Nov 2026	Closed Book
Comprehensive Exam	180 minutes	40%	80	07 Dec 2026 (FN)	Closed Book

Best two quizzes out of three would be considered for the final evaluation. All quizzes are announced.

* Quiz Dates are tentative and subject to change as per the circumstances. In case of any change, notice will be published in advance.



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Criterion for NC / Minimum passing marks criterion / Minimum performance requirement for a valid grade: A student should obtain 30 % of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

7. ATTENDANCE POLICY

Each student is expected to attend all classes and participate in all evaluation components. Discussions during class hours will be highly beneficial for better learning and higher scores. **The AUGSD's attendance policy will be strictly followed.**

8. CHAMBER CONSULTATION HOUR

Students can approach the Instructor In-Charge (IC) at any time via email. Their queries will be answered via email or in offline meetings, depending on the level of the question. The instructors will also be available to meet each Tuesday from 4 PM to 5 PM in their chambers (Prof. Balakrushna Padhi, 6165-C, and Prof. Rahul Arora, 6165-D) in the New Academic Block (NAB).

9. MAKEUP POLICY

Makeup for the mid-semester exam and comprehensive examination shall be granted only in genuine cases, as per the academic regulations, after a thorough examination of the makeup application. Only those requests would be entertained for a thorough examination who applied **before or within two days of the exam**. Please do not make requests after the application period is over. Those requests would not be entertained. Also, **no makeup request for the quizzes containing one buffer shall be considered.**

10. NOTICES

For course-related announcements, please check your email and LMS (NALANDA) accounts regularly. All the Information will also be communicated during the class.

11. NOTE

Quizzes and exams will be conducted during scheduled hours as specified. Changes, if any, will be communicated well in advance. Please check/write your ID number and name before submitting any evaluative component/request for evaluation. Failure to do so may result in unevaluated answer sheets.

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